

# Sanidhya Singh

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## Professional Summary

AI/ML Engineer with 2+ years of experience designing and shipping production-grade AI systems across life sciences, MLOps, computer vision, and NLP. Spearheaded a **LangGraph multi-agent clinical trial verification platform** for a life sciences client — taking it from architecture to production independently. Specialise in **Generative AI, RAG pipelines, and multi-agent orchestration**, with a consistent track record of owning complex problems end-to-end and delivering them into real environments. Immediate joiner, open to remote and on-site roles.

## Technical Skills

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|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GenAI & Agents     | LangChain, LangGraph, RAG, MCP Servers, Multi-Agent Systems, Prompt Engineering, Fine-tuning (LoRA/PEFT), LLM Evaluation (RAGAS), OpenAI API, Hugging Face |
| ML & Deep Learning | PyTorch, TensorFlow, Scikit-learn, Transformers, NLP, Embeddings, CNN, Pandas, NumPy                                                                       |
| Computer Vision    | YOLOv3, DeepSORT, MTCNN, TensorRT, OpenCV                                                                                                                  |
| Backend & APIs     | Python (Async, Pydantic), FastAPI, REST APIs, Microservices, Celery, Redis, SQL                                                                            |
| MLOps & Cloud      | GCP, AWS S3, Azure AI, Docker, Kubernetes, MLflow, Apache Beam, Dataiku, Git                                                                               |
| Databases          | PostgreSQL, ChromaDB, Pinecone, FAISS, NoSQL                                                                                                               |

## Professional Experience

**AI/ML Engineer** | *MoreYeas IT* Nov 2025 – Mar 2026  
Indore, Madhya Pradesh

- Spearheaded end-to-end delivery of a **GenAI-powered clinical trial data verification platform** for a life sciences client — sole owner of the full lifecycle from requirement gathering and system design through to production deployment.
- Drove critical architectural decisions independently: selected **LangGraph** for multi-agent orchestration, designed the RAG retrieval and chunking strategy, and structured async **FastAPI** microservices with **Celery** task queues for concurrent document batch processing.
- Built an OCR + NER document intelligence layer for clinical fact extraction with full evidence traceability, and integrated **MCP servers** for modular tool exposure across agents.

**Jr. Data Scientist** | *MoreYeas Technologies* Jun 2022 – Apr 2024  
Indore, Madhya Pradesh

- Took full ownership of **Sentry-B** — a real-time edge AI surveillance system — designing the complete pipeline from live CCTV ingestion through person detection (YOLOv3), persistent tracking (DeepSORT), face extraction (MTCNN), and identity verification, deployed on Jetson Nano.
- Architected a **proactive MLOps failure-detection pipeline** for Dish/Sling TV — designed the ETL strategy in Dataiku, engineered an Isolation Forest anomaly detection model on server logs, and deployed MLflow-tracked experiments with automated alerting dashboards.
- Delivered **OCR-based financial document pipelines** and LLM-powered NLP applications as FastAPI microservices on GCP, iterating directly with clients to improve extraction accuracy.

## Key Projects

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### GenAI Clinical Trial Source Data Verification System

LangChain · LangGraph · RAG · MCP Servers · OCR · NER · FastAPI · Celery · PostgreSQL

- Eliminated manual cross-checking for a life sciences team by building a **LangGraph multi-agent pipeline** that automatically maps source documents to EDC records in CDISC ODM XML format — replacing a fully human-driven clinical data verification process.
- Engineered a RAG-based document intelligence layer combining OCR and NER to extract clinical facts from PDFs and link them to structured database records with full evidence traceability — enabling auditable, source-traceable verification at scale.
- Built a mismatch detection engine to surface discrepancies between extracted clinical facts and database entries; integrated **MCP servers** for modular, composable tool exposure across agents.
- Designed the delivery architecture as async **FastAPI** microservices with Celery task queues, built for concurrent processing of large clinical trial document batches in a production environment.

### MLOps Log Intelligence Pipeline — Dish/Sling TV

Python · Dataiku · AWS S3 · Apache Beam · Scikit-learn · GCP · Docker · MLflow

- Replaced reactive manual log monitoring for Dish/Sling TV by architecting an end-to-end **proactive failure-detection pipeline** — ingesting and analysing logs from **10 critical infrastructure machines** via a Dataiku-based ETL strategy with a structured data lake on AWS S3.
- Trained an **Isolation Forest** anomaly detection model with time-series feature engineering to predict machine failures before they occurred — giving ops teams advance warning to intervene before downtime.
- Deployed **MLflow** experiment tracking and automated alerting dashboards, giving ops teams full proactive failure visibility and eliminating the need to manually inspect raw log files.

### Sentry-B: Intelligent Edge Surveillance & Attendance System

YOLOv3 · DeepSORT · MTCNN · PyTorch · TensorRT · Jetson Nano · OpenCV

- Deployed a real-time edge AI surveillance system on **Jetson Nano** capable of detecting, tracking, and face-identifying personnel from live CCTV feeds — built on a YOLOv3 + DeepSORT + MTCNN pipeline with automated employee vs. guest classification.
- Achieved **25+ FPS** real-time inference on constrained edge hardware by optimising the full model stack with **TensorRT** — carefully balancing precision and latency trade-offs across each model in the pipeline.
- Extended the system with an emotion detection module and automated daily attendance marking, adding workforce sentiment monitoring on top of the existing identity verification pipeline.

### AI-Powered Resume Optimization Engine

Python · LangChain · ChromaDB · FastAPI · Pydantic · OpenAI API · Embeddings

- Built a **RAG-based resume evaluation tool** using LangChain and ChromaDB that semantically compares resumes against job descriptions — surfacing genuine skill gaps rather than surface-level keyword mismatches, enabling smarter, targeted job applications.
- Designed a custom **ATS scoring engine** that jointly evaluates keyword alignment, skill relevance, and experience context — served via async FastAPI endpoints with Pydantic-enforced structured LLM output for consistent, parseable scoring results.

## Education

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M.Tech in Data Science | DAVV University, School of Data Science & Forecasting

Indore · May 2026

B.Tech in Information Technology | Medi-Caps University

Indore · Jun 2023

## Certifications

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The MCP Course: Unit 1 — Fundamentals of MCP | Hugging Face

Apr 2026

Microsoft Azure AI Essentials: Workloads & ML on Azure | LinkedIn Learning

Apr 2026